

## Savitribai Phule Pune University

## T.Y.B.Sc.(C.S.) (NEP – 2020) SEM – V Practical Examination

## Lab Course CS-306-MJ-P

## Core Java and Web Technology-I

Duration: 3 Hours

Maximum Marks:35

**Q1)** Write a Java program to print the sum of elements of the array. Also [15 marks]  
display array elements in ascending order.

**Q2)** Design a College/School Registration Form using HTML5 and CSS3 and [15 marks]  
apply appropriate HTML5 validation attributes.

**My-Future College / School**  
Please Fill Below Detail In Student Registration Form

Student name :  Min 3 & Max 10 letter is allow

Date of birth :  Valid Date is allow

Mobile no. :  Only integer is allow

Email id :  Valid Email Id is allow

Password :  Min 8 digit is allow

Gender : ☐ Male ☐ Female

Hobbies : ☐ Drawing ☐ Singing ☐ Dancing ☐ Sketch ☐ Sports

Course :  ▼

Student photo :  No file chosen

Register Time:  ⌚

Address :

☐ I hereby declare that above information provided is correct

**Q3)** Viva [05 marks]

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**Q1)** Write a Java Program to Display Armstrong Numbers Between range. [15 marks]  
Accept range from user.

**Q2)** Design a Student Registration Form using HTML5 and CSS3 and apply appropriate HTML5 validation attributes. [15 marks]

**Student Registration Form**

Roll no. :

Student name :  -

Father's name :

Date of birth :  -  -  (DD-MM-YYYY)

Mobile no. :  -

Email id :

Password :

Gender : ☐ Male ☐ Female

Department : ☐ CSE ☐ IT ☐ ECE ☐ Civil ☐ Mech

Course :

Student photo :  No file chosen

City :

Address :

**Q3)** Viva [05 marks]

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- 
- Q1)** Write a package for String operation which has two classes Con and Comp. Con class has to concatenate two strings and comp class compares two strings. Also display proper messages on execution. **[15 marks]**
- Q2)** Create a responsive webpage using CSS Flexbox for a college homepage containing header, navigation, content and footer. Use appropriate Flexbox alignment and distribution properties **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a menu driven program to perform the following operations on multidimensional array ie matrix : **[15 marks]**
- i. Addition
  - ii. Multiplication
  - iii. Transpose of any matrix.
  - iv. Exit
- Q2)** Create a webpage containing a heading and a button. When the button is clicked, change the heading text from “Hello! Welcome” to “Text Changed Successfully!”. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- |            |                                                                                                                          |                   |
|------------|--------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Write a program to accept a text file from user and display the contents of a file in reverse order and change its case. | <b>[15 marks]</b> |
| <b>Q2)</b> | Create an image gallery using HTML5 and CSS3 with hover effects, responsive images and Flexbox/Grid layout.              | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                     | <b>[05 marks]</b> |

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- 
- Q1)** Write a program to define a class Account having members custname, accno. Define default and parameterized constructor. Create a subclass called SavingAccount with members savingbal, minbal. Create a derived class AccountDetail that extends the class SavingAccount with members, depositamt and withdrawamt. Write a appropriate method to display customer details. **[15 marks]**
- Q2)** Create a webpage containing a button and link. Change their appearance when the mouse pointer moves over them. (Use :hover Pseudo-Class) **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a class Driver with attributes license\_no, name, address and age. Initialize values through the parameterized constructor. If age of Driver is less than 18 then user-defined exception should be generated —Age is below 18 years— **[15 marks]**
- Q2)** Create an unordered list of Programming Language names and apply different styles to the first and last list items.(Use :first-child and :last-child) **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contain only the method printArea() that prints the area of the given shape. (use method overriding). **[15 marks]**
- Q2)** Create a webpage using JavaScript DOM manipulation and event handling with an image and a button. When the button is clicked, replace the displayed image with another image. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a program which define class Employee with data member as id, name and salary Store the information of 'n' employees and display the name of employee having maximum salary (Use array of object) **[15 marks]**
- Q2)** Write a JavaScript program to demonstrate Arrow Functions for addition, subtraction, multiplication and division. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- |            |                                                                                                                                                |                   |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Write a program to read the contents of “abc.txt” file. Display the contents of file in uppercase as output.                                   | <b>[15 marks]</b> |
| <b>Q2)</b> | Write a JavaScript program to store student information and marks in variables and generate a formatted student report using template literals | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                                           | <b>[05 marks]</b> |

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- 
- |            |                                                                                                                                                                                                                                                                                                                                            |                   |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Write a program to create a super class Vehicle having members Company and price. Derive two different classes LightMotorVehicle (mileage) and HeavyMotorVehicle (capacity_in_tons). Accept the information for "n" vehicles and display the information in appropriate form. While taking data, ask user about the type of vehicle first. | <b>[15 marks]</b> |
| <b>Q2)</b> | Write a JavaScript program using an arrow function to calculate the total and average marks of three subjects. Display the result using template literals                                                                                                                                                                                  | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                                                                                                                                                                                                                                       | <b>[05 marks]</b> |

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- 
- Q1)** Write a package game which will have 2 classes Indoor & Outdoor. Use a function display () to generate the list of players for the specific game. Use default & parameterized constructor **[15 marks]**
- Q2)** Write a JavaScript program using an arrow function to calculate the total price of a product. Display the bill using template literals and string interpolation. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Define a “Clock” class that does the following; **[15 marks]**
- a. Accept Hours, Minutes and Seconds
  - b. Check the validity of numbers
  - c. Set the time to AM/PM mode
- Use the necessary constructors and methods to do the above task
- Q2)** Write a JavaScript program to swap two numbers without using a third variable using array destructuring **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a JavaFX program that takes a numeric input from a user via a TextField and displays Whether it is a prime number in an output field upon clicking a “Process” button. **[15 marks]**
- Q2)** Create an object containing employee details such as name, department, and salary. Use object destructuring to extract and display the values **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a JavaFX program to build a simple calculator with digit buttons (0-9) and basic arithmetic operator buttons (+, -, \*, /). Implement action listeners for each button (digits and operators) to perform calculations. **[15 marks]**
- Q2)** Write a JavaScript program to create a copy of an existing array using the spread operator and add a new element to the copied array. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- |            |                                                                                                                                                            |                   |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Write a JavaFX application that simulates a shopping cart. Add items with buttons, and implement action listeners to update the cart when an item is added | <b>[15 marks]</b> |
| <b>Q2)</b> | Write a JavaScript program using a rest parameter to accept any number of values and calculate their sum                                                   | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                                                       | <b>[05 marks]</b> |

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- 
- |            |                                                                                                                                                                      |                   |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Write a JavaFX application that changes the background color of a window based on a combination of keys pressed (e.g., Ctrl+Alt+O changes the background to orange). | <b>[15 marks]</b> |
| <b>Q2)</b> | Create a JavaScript module containing an arithmetic operation functions. Export the function and import it into another JavaScript file to perform all operations.   | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                                                                 | <b>[05 marks]</b> |

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- |            |                                                                                                                                                       |                   |
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| <b>Q1)</b> | Creating an application with multiple buttons (e.g., Red, Green, Blue) that print their respective color or change the background color when clicked. | <b>[15 marks]</b> |
| <b>Q2)</b> | Create a Node.js HTTP server that sends an appropriate response to a client request.                                                                  | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                                                  | <b>[05 marks]</b> |

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- |            |                                                                                                                                                                                       |                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Write a Java program to accept a file name from the user and count the number of characters, words, and lines present in the file.                                                    | <b>[15 marks]</b> |
| <b>Q2)</b> | Write a JavaScript program using async/await to simulate user login. If the username and password are correct, display a success message; otherwise, handle the error using try-catch | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                                                                                  | <b>[05 marks]</b> |

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- 
- Q1)** Design a screen to handle the Mouse Events such as MOUSE\_MOVED and MOUSE\_CLICK and display the position of the Mouse\_Click in a TextField. **[15 marks]**
- Q2)** Write a JavaScript program using a Promise to divide two numbers. If the denominator is zero, reject the Promise and handle the error. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a program to create parent class College (cno, cname, caddr) and derived class Department (dno, dname) from College. Write a necessary methods to display College details. **[15 marks]**
- Q2)** Write a Node.js program using the path module to perform operations such as joining, resolving and extracting file/directory information. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a program using interface, create a class Product (product\_id, product\_name, product\_cost, product\_quantity) default and parameterized constructor. Create objects of class product and display the contents of each object and Also display the object count. **[15 marks]**
- Q2)** Write a Node.js program using the fs module to perform file operations asynchronously. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a program for multilevel inheritance such that Country is inherited from Continent. State is inherited from Country. Display the place, State, Country and Continent. **[15 marks]**
- Q2)** Write a Node.js program to create, write, read and append data to a file using the fs module. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- |            |                                                                                                                                |                   |
|------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Create an abstract class Shape with methods area & volume. Derive a class Cylinder (radius, height). Calculate area and volume | <b>[15 marks]</b> |
| <b>Q2)</b> | Write a Node.js program to demonstrate synchronous and asynchronous file operations and compare their execution.               | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                           | <b>[05 marks]</b> |

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- 
- |            |                                                                                                                                                                                                  |                   |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Write a Java program that defines an interface Calculator containing methods add() and subtract() and implements it in class SimpleCalc.                                                         | <b>[15 marks]</b> |
| <b>Q2)</b> | Write a Node.js program to perform directory management operations such as creating, listing, renaming and deleting directories. Also demonstrate JSON parsing and writing structured JSON data. | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                                                                                             | <b>[05 marks]</b> |

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- 
- Q1)** Define a class Student with data members student\_name, student\_rollno, total\_lectures, attended\_lectures. Create an object of Student. If attendance percentage is less than 75%, throw a user-defined exception “Student is Not Eligible for Exam”, otherwise display student details. **[15 marks]**
- Q2)** Write a Node.js program using the fs module to copy, rename and delete files asynchronously. Display appropriate success and error messages. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Write a program to define class Person with data members as Personname,Aadharno, Panno. Accept information for 5 objects and display appropriate information (use this keyword). **[15 marks]**
- Q2)** Write a Node.js program to read a JSON file, modify selected properties of the JSON object, and write the updated JSON data back to the file using the fs module **[15 marks]**
- Q3)** Viva **[05 marks]**

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- 
- Q1)** Define a class MyDate(Day, Month, year) with methods to accept and display a MyDateobject. Accept date as dd,mm,yyyy. Throw user defined exception "InvalidDateException" if the date is invalid. **[15 marks]**
- Q2)** Create a Node.js program to list files in a specified directory, filter files based on their extensions (such as .txt, .json and .js), and display the matching file names. **[15 marks]**
- Q3)** Viva **[05 marks]**

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- |            |                                                                                                                                                                                                        |                   |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Q1)</b> | Write a program to accept a number from the user, if number is zero then throw user defined exception “Number is 0” otherwise check whether no is prime or not (Use static keyword)                    | <b>[15 marks]</b> |
| <b>Q2)</b> | Write a Node.js program to demonstrate file statistics using fs.stat(), including file size, file type, creation/access/modification information, and whether the path represents a file or directory. | <b>[15 marks]</b> |
| <b>Q3)</b> | Viva                                                                                                                                                                                                   | <b>[05 marks]</b> |

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- 
- Q1)** Define a class MyNumber having one private integer data member. Write a default constructor initialize it to 0 and another constructor to initialize it to a value. Write methods isNegative, isPositive, isOdd, isEven. Use command line argument to pass a value to the object and perform the above operations. **[15 marks]**
- Q2)** Create a Node.js mini file-management application using the fs and path modules that allows the user to create, read, update, rename and delete text files through a menu-driven interface. **[15 marks]**
- Q3)** Viva **[05 marks]**